7.5 Volume of Cuboid and Cube

All of us have tasted 50 *ml* and 100 *ml* of ice cream. Take one such 100 *ml* ice cream cup. This cup can contain 100 *ml* of water, which means that the capacity or volume of that cup is 100 *ml*. Take a 100 *ml* cup and find out how many such cups of water can fill a jug. If 10 such 100 *ml* cups can fill a jug then the capacity or volume of the jug is 1 litre $\left(10 \times 100\, ml = 1000\, ml = 1l\right)$. Further check how many such jug of water can fill a bucket. That is the capacity or volume of the bucket. Likewise we can calculate the volume or capacity of any such things.

Volume is the measure of the amount of space occupied by a three dimensional solid. Cubic centimetres (cm^3) , cubic metres (m^3) are some cubic units to measure volume.

Note

Unit Cube :

A cube with side 1 unit.

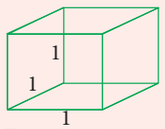
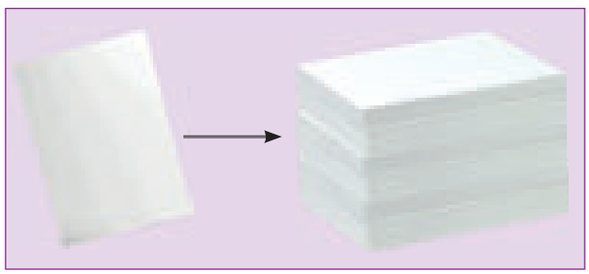


Fig. 7.17

Volume of the solid is the product of 'base area' and 'height'. This can easily be understood from a practical situation. You might have seen the bundles of A4 size paper. Each paper is rectangular in shape and has an area (=*lb*). When you pile them up, it becomes a bundle in the form of a cuboid; *h* times *lb* make the cuboid.


Fig. 7.18

7.5.1 Volume of a Cuboid

Let the length, breadth and height of a cuboid be *l*, *b* and *h* respectively.

Then, volume of the cuboid

$$V = (\text{cuboid's base area}) \times \text{height}$$
$$= (l \times b) \times h = lbh \text{ cubic units}$$

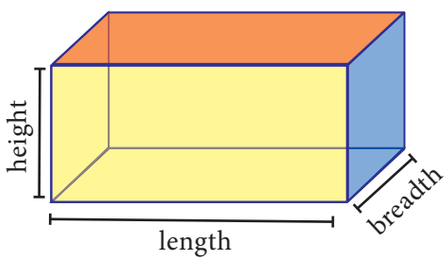


Fig. 7.19

Note

The units of length, breadth and height should be same while calculating the volume of a cuboid.

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